

DTSA 5021

Statistical Learning for Data Science: Resampling, Selection and Splines

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 3 – Core	Machine Learning Engineer (39.9 / 100)	28.7 / 100
SPECIALIZATION Intro to Machine Learning: Supervised Learning (Statistical Learning specialization)		

29	54	+42
CURRENT AVG SCORE	POTENTIAL SCORE	MAX SCORE GAIN
across 6 job roles	with recommended additions	best single-role improvement

CURRENT COURSE DESCRIPTION

Consists of the foundational framework & application of cross-validation, bootstrapping, dimensionality reduction, ridge regression, lasso, GAMs and splines.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	39.9	57.1	+17.2	MEDIUM
MLOps Engineer	30.0	71.6	+41.6	HIGH
Data Scientist	28.5	53.5	+25.0	MEDIUM
ML Engineer	26.9	44.9	+17.9	MEDIUM
AI Engineer	23.8	50.1	+26.3	MEDIUM
Applied Scientist	23.0	48.8	+25.9	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design

4 of 6 roles

systems

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

HIGH

Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.

GAP KEYWORDS BY ROLE

Machine Learning Engineer	39.9 → 57.1 (+17.2 pts)
data learning tools science models model applications machine	
MLOps Engineer	30.0 → 71.6 (+41.6 pts)
data model scientists text systems learning machine tools	
Data Scientist	28.5 → 53.5 (+25.0 pts)
data product problems	
ML Engineer	26.9 → 44.9 (+17.9 pts)
data learning machine systems models model world problems	
AI Engineer	23.8 → 50.1 (+26.3 pts)
data models learning model systems problems machine information	
Applied Scientist	23.0 → 48.8 (+25.9 pts)
computational problems models data learning systems industry world	

Methodology:

Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.